AWS NETWORKING · EWB CLOUD LABS PROJECT 01 · CASE STUDY

AWS VPC Peering: Private Connectivity, Route Validation, and Security Testing

Two VPCs, one peering connection, a bastion host, route validation, security-group restriction, controlled failure, recovery and cleanup.



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AWS REGION	Asia Pacific (Mumbai) · ap-south-1
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Historical lab demonstration. The AWS environment shown in this report has been cleaned up; it is not running.

Submission file: Abdul_Aziz_EWB_VPC_Peering.pdf

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1. Executive Summary

This project connected two Amazon VPCs in the Mumbai Region (**ap-south-1**) with a VPC peering connection so that two private EC2 instances could exchange traffic using only their private IPv4 addresses, while a single public bastion host in VPC A provided SSH administration. It then tested the design: SSH through the bastion, ICMP in both directions, and a deliberate ICMP failure caused by a security-group change.

The report separates three things throughout: the **historical lab configuration** (what was built), the **observed test state** (what the supplied screenshots show), and the **post-cleanup state** (every project resource was deleted afterwards, as reported by the author). The AWS environment described here is not running.

Table 1 — Project at a glance

Item	Result
Topology	VPC A 10.10.0.0/16 (public + private subnet), VPC B 10.20.0.0/16 (private subnet), one peering connection, one Internet gateway on VPC A only
Instances	ewb-bastion 10.10.1.10 (public [redacted public IPv4]), ewb-app-a 10.10.2.10, ewb-app-b 10.20.1.10; all t3.micro, ap-south-1a
Peering	ewb-a-b-peer, pcx-029955b9450e10c12, Active during testing (E12); deleted afterwards (reported)
Verified by screenshot	Peering Active; both private-subnet route tables with peering routes; Internet gateway attached to VPC A; App-A private-only and App-B private-only; SSH to bastion, App-A and App-B; App-B to App-A ping 4/4, 0% loss; App-A to App-B ping 100% loss during the controlled failure
Reported results and actions	Baseline failure before peering; App-A to App-B success; recovery after restoring the rule; cleanup actions
Verification notes	Screenshot-supported results and reported outcomes are distinguished in the test summary and evidence index.

2. Project Objectives and Scope

- Plan non-overlapping VPC and subnet CIDRs and place each workload in an explicit subnet with an explicit route-table association.
- Peer VPC A and VPC B, add routes on **both** sides and permit only the required inbound traffic with source-restricted security-group rules.
- Keep both application servers without public IPv4 addresses and administer them through one public bastion.
- Show that connectivity works in both directions, then show a controlled failure and recovery, then clean up.

Scope boundaries

This is a small teaching lab in one account, one Region and one Availability Zone. It does not demonstrate high availability, disaster recovery, production-grade security, performance, DNS resolution over peering, cross-account or cross-Region peering, or transitive routing. VPC B deliberately has no Internet gateway and no NAT gateway.

3. Requirements and Success Criteria

Requirements come from the EWB project brief (Attachment A, 14 pages). Acceptance tests are those on brief page 11; deliverables are those on page 13. Evidence coverage for each requirement is in Appendix A.

Table 2 — Requirements extracted from the EWB brief

#	Requirement (brief page)	Success criterion
1	Non-overlapping CIDRs (p.4)	10.10.0.0/16 and 10.20.0.0/16; each /24 inside its own VPC
2	Public and private subnets with explicit route-table associations (p.5–6)	ewb-a-public ↔ ewb-rt-a-public, ewb-a-private ↔ ewb-rt-a-private, ewb-b-private ↔ ewb-rt-b-private
3	Internet gateway on VPC A only (p.5–6)	ewb-igw-a attached to ewb-vpc-a; default route only in the public route table
4	Peering Active with routes on both sides (p.9–10)	Peering Active; peer routes in all route tables that need them
5	Least-privilege inbound rules (p.6)	Bastion SSH from admin /32; app SSH from bastion /32; ICMP from the peer /32
6	Private instances stay private (p.7, 11)	No public IPv4 on App-A or App-B
7	Test 01–05 and controlled failure (p.11)	Baseline no replies; 4/4 replies each way; App-B SSH via bastion; replies stop when the rule is removed and resume when restored
8	Deliverables (p.13)	Cover, diagram + inventory, subnets + 3 route tables, peering + SG rules, tests, failure/recovery, cleanup, learning note; file Abdul_Aziz_EWB_VPC_Peering.pdf; no keys or tokens
9	Cleanup (p.14)	Instances, peering, route tables, subnets, IGW and VPCs removed; evidence captured

4. Architecture Overview and Diagram

Reading the architecture

The administrator reaches the **bastion** (public subnet of VPC A) through the Internet gateway; the bastion is the only host with a public IPv4 address. From the bastion, SSH reaches App-A over VPC A's local route and App-B across the peering connection. App-A and App-B talk directly over private IPs, via the peering routes. VPC B has no Internet gateway, so peering gives it private reach to VPC A but no path to the internet through VPC A.

Diagram scope

The architecture diagram was rebuilt around the documented resource inventory and available screenshots. Items based on the project brief rather than a dedicated console capture are identified in the evidence index; no unobserved resources are represented as verified.

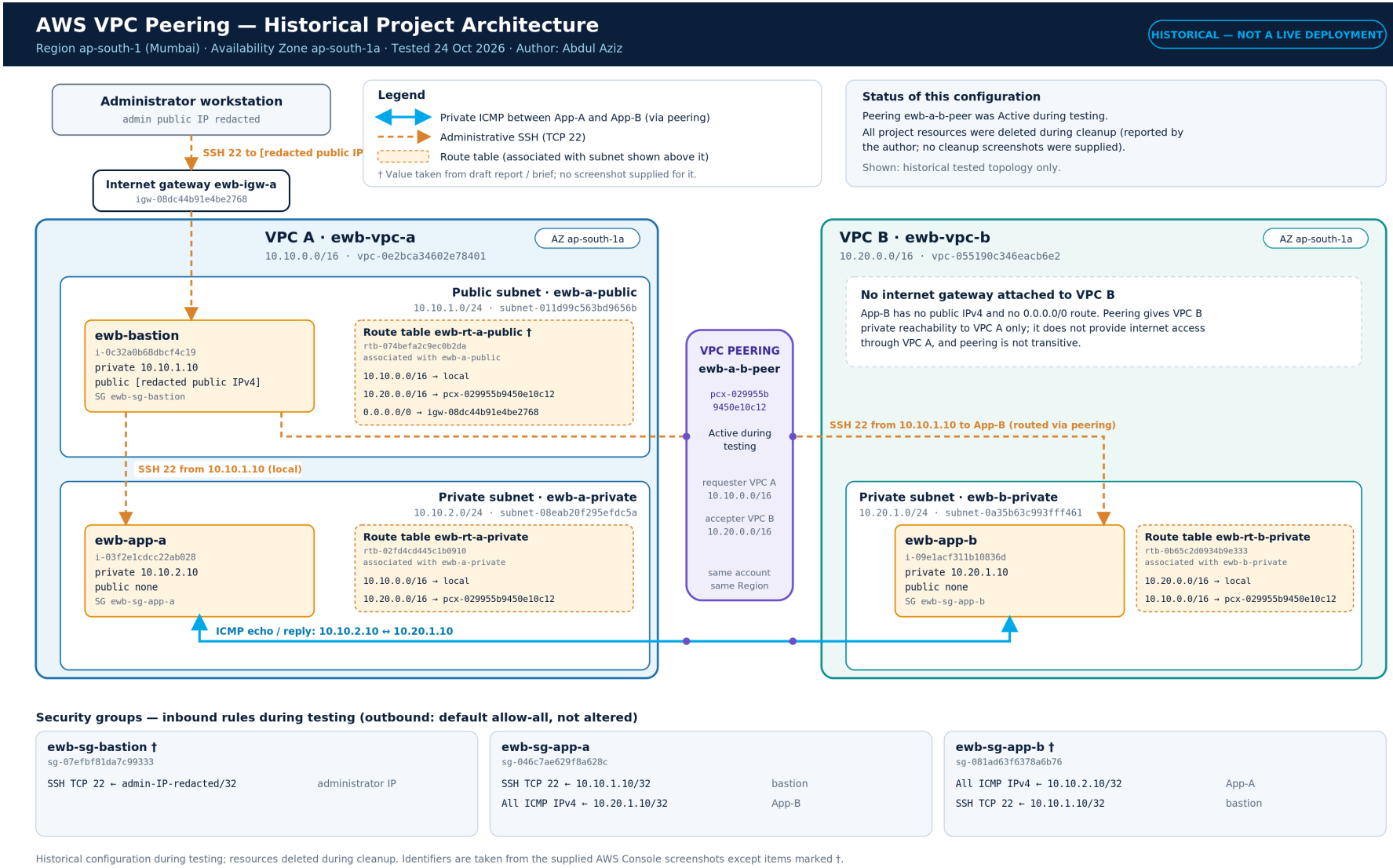


Figure 1 — Historical project architecture (ap-south-1a, tested 24 Oct 2026)
Shows: both VPCs with CIDRs, subnets, instances, route tables, peering, IGW, SSH and ICMP flows, security-group rules
Demonstrates: the verified topology; † marks values from the draft/brief with no supporting screenshot

5. CIDR Allocation and Subnet Design

Table 3 — VPCs

VPC	Name	VPC ID	IPv4 CIDR	Evidence
VPC A	ewb-vpc-a	vpc-0e2bca34602e78401	10.10.0.0/16	ID: E05,E06,E08,E12; CIDR: E12
VPC B	ewb-vpc-b	vpc-055190c346eacb6e2	10.20.0.0/16	ID: E07,E12; CIDR: E12

Table 4 — Subnets and route-table associations

Subnet	Subnet ID	CIDR	Type	AZ	Associated resource	Route table
ewb-a-public	subnet-011d99c563bd9656b	10.10.1.0/24	Public	ap-south-1a	ewb-bastion	ewb-rt-a-public
ewb-a-private	subnet-08eab20f295efdc5a	10.10.2.0/24	Private	ap-south-1a	ewb-app-a	ewb-rt-a-private
ewb-b-private	subnet-0a35b63c993fff461	10.20.1.0/24	Private	ap-south-1a	ewb-app-b	ewb-rt-b-private

Why these ranges. 10.10.0.0/16 and 10.20.0.0/16 do not overlap, which AWS requires before two VPCs can be peered; two non-overlapping *subnets* inside overlapping VPC ranges would not fix this. Each /24 sits inside its own VPC range and provides 251 usable addresses, ample for a three-host lab. All three subnets are in ap-south-1a (E05–E07), which keeps the lab simple and avoids cross-AZ considerations; it also means the design has a single-AZ failure domain.

Evidence note. Subnet names, IDs and AZ come from E05–E07. The CIDR blocks themselves were not captured from the Subnets console; E02 shows `10.10.2.10/24` on App-A, consistent with 10.10.2.0/24, and the others rely on the draft and brief.

6. Resource Inventory

Table 5 — VPC peering connection

Peering	ID	Requester	Accepter	Scope	During testing	After cleanup
ewb-a-b-peer	pcx-029955b9450e10c12	vpc-0e2bca34602e78401 / ewb-vpc-a (10.10.0.0/16)	vpc-055190c346eacb6e2 / ewb-vpc-b (10.20.0.0/16)	Mumbai (ap-south-1) both sides; same account	Active (E12)	Deleted (reported by author; no screenshot)

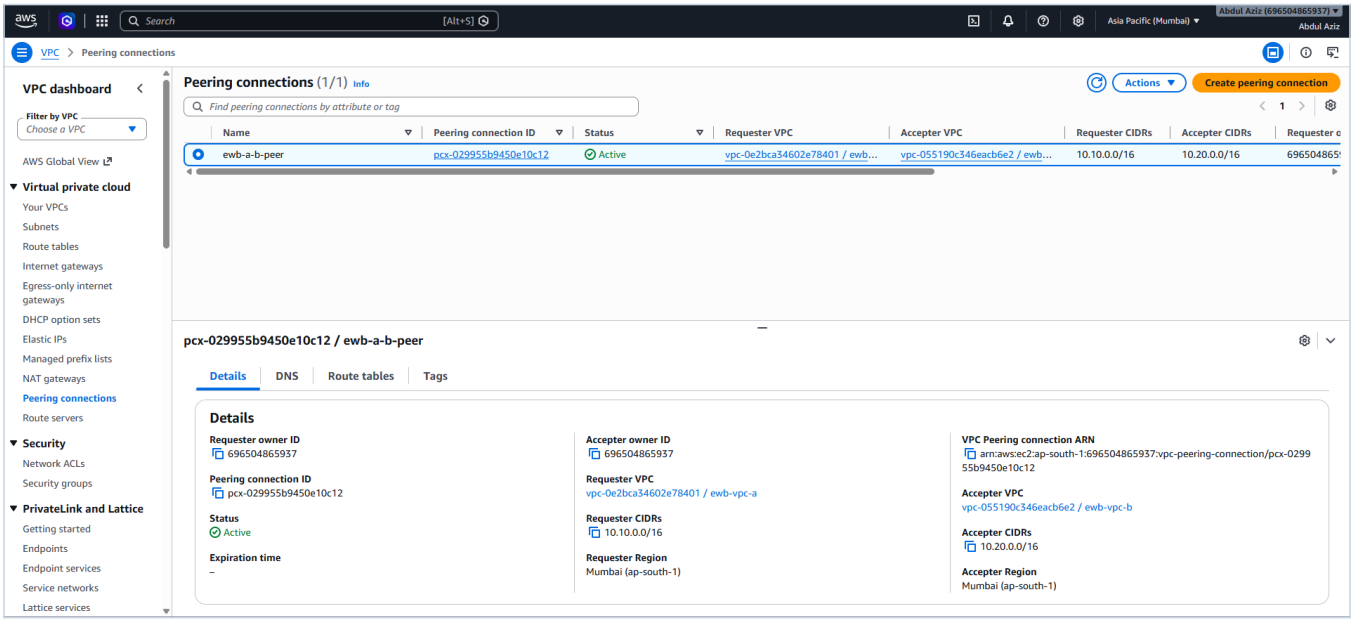


Figure 2 — VPC Peering Status: Active During Testing [E12]
Observed: pcx-029955b9450e10c12 / ewb-a-b-peer, Status Active; requester vpc-0e2bca34602e78401 (10.10.0.0/16); accepter vpc-055190c346eacb6e2 (10.20.0.0/16); both Mumbai
Demonstrates: the link exists between the correct VPCs; Active status alone does not prove reachability

Table 6 — EC2 instances (all terminated after testing — reported)

Instance	Instance ID	Type	Private IP	Public IPv4	Subnet	Security group
ewb-bastion	i-0c32a0b68dbcf4c19	t3.micro	10.10.1.10	[redacted public IPv4] (auto-assigned)	ewb-a-public	ewb-sg-bastion sg-07efbf81da7c99333
ewb-app-a	i-03f2e1cdcc22ab028	t3.micro	10.10.2.10	none	ewb-a-private	ewb-sg-app-a sg-046c7ae629f8a628c
ewb-app-b	i-09e1acf311b10836d	t3.micro	10.20.1.10	none	ewb-b-private	ewb-sg-app-b sg-081ad63f6378a6b76

The security-group names and IDs are listed by the console in E13 (ewb-sg-app-a and ewb-sg-bastion in VPC A, ewb-sg-app-b in VPC B). Which group each instance used was not visible in the instance screenshots; the association is taken from the draft and brief.

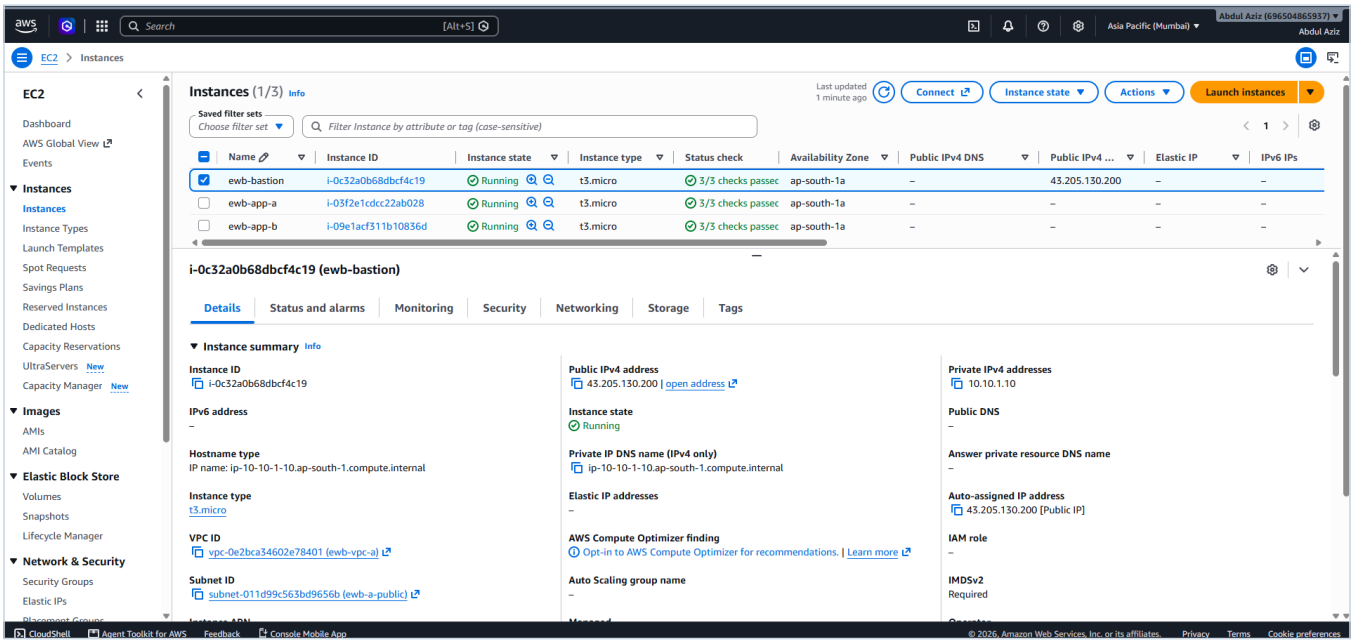


Figure 3 — EC2 Inventory: ewb-bastion [E05]
Observed: three instances Running, 3/3 status checks, t3.micro, ap-south-1a; bastion selected: public IPv4 43.205.130.200, private 10.10.1.10, subnet ewb-a-public in ewb-vpc-a
Demonstrates: the bastion is the only instance with a public IPv4 address

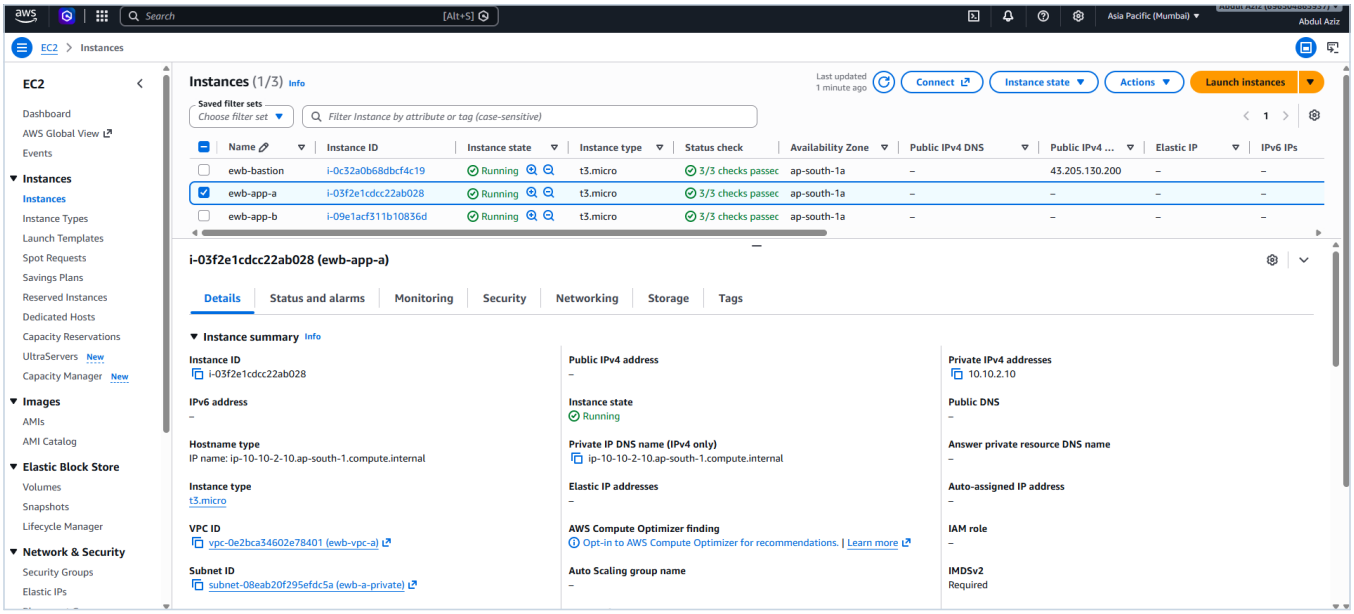


Figure 4 — EC2 Inventory: ewb-app-a (Private Only) [E06]
Observed: i-03f2e1cdcc22ab028, Public IPv4 address '-', private 10.10.2.10, subnet ewb-a-private in ewb-vpc-a
Demonstrates: App-A has no public IPv4 address

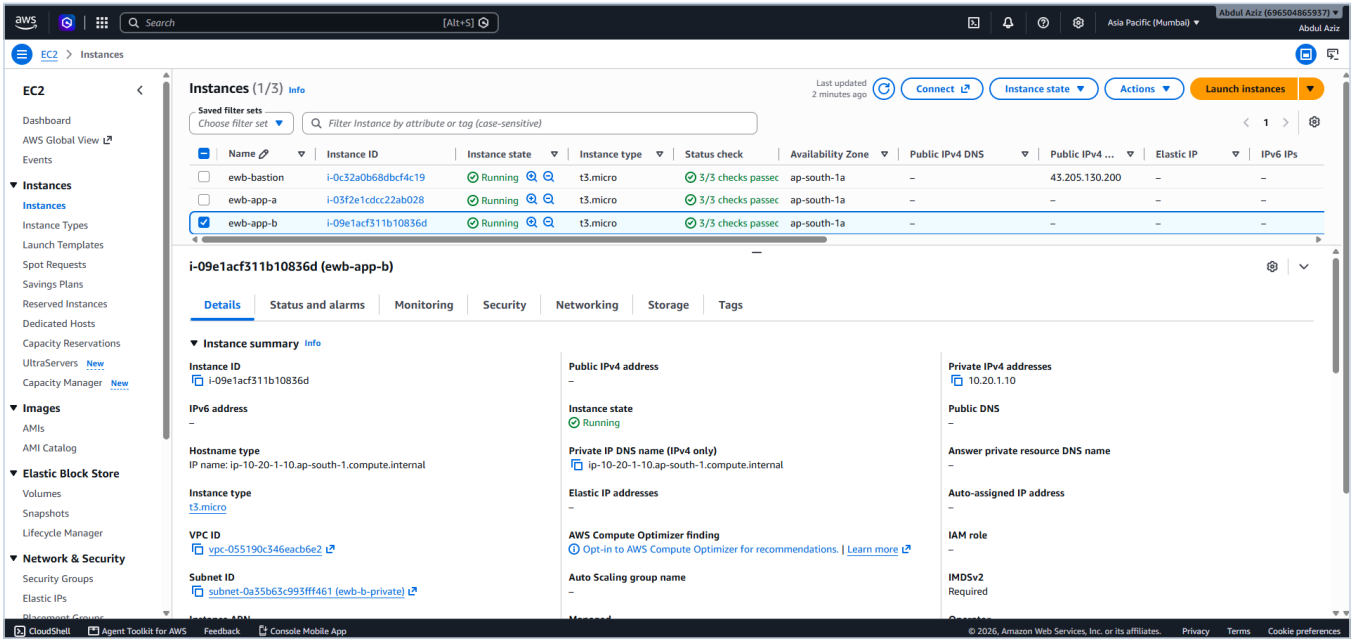


Figure 5 — EC2 Inventory: ewb-app-b (Private Only) [E07]
Observed: i-09e1acf311b10836d, Public IPv4 address '-', private 10.20.1.10, VPC vpc-055190c346eacb6e2, subnet ewb-b-private
Demonstrates: App-B has no public IPv4 address and sits in VPC B

Table 7 — Internet gateway

Internet gateway	ID	Attached to	State	After cleanup
ewb-igw-a	`igw-08dc44b91e4be2768`	vpc-0e2bca34602e78401 / ewb-vpc-a	Attached (E08)	Deleted (reported by author; no screenshot)

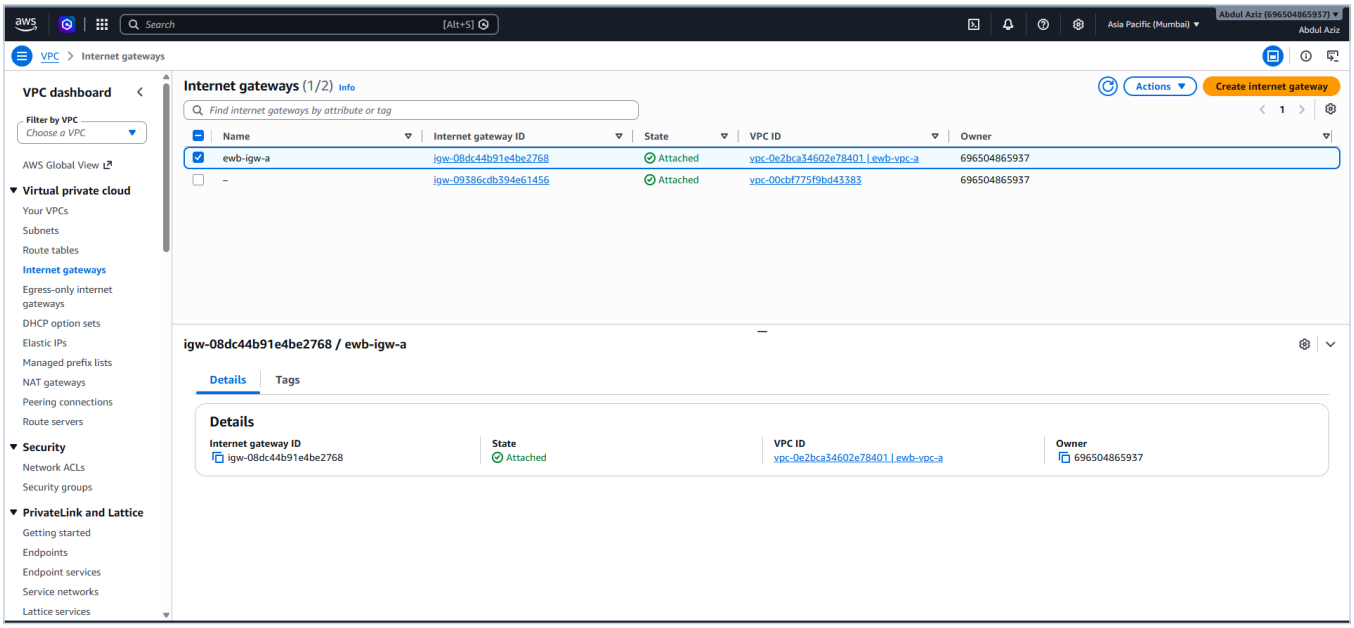


Figure 6 — Internet Gateway ewb-igw-a Attached to VPC A [E08]
Observed: ewb-igw-a igw-08dc44b91e4be2768 Attached to vpc-0e2bca34602e78401 (ewb-vpc-a); a second IGW igw-09386cdb394e61456 is attached to the account's default VPC
Demonstrates: VPC A has the Internet gateway; no IGW is attached to VPC B

7. Route-Table Configurations and Subnet Associations

A subnet uses the route table it is explicitly associated with, and traffic leaves the subnet using the **most specific** matching route. Local routes cover each VPC's own CIDR; the peering routes send the peer VPC's CIDR to the peering connection. The public table also has a default route to the Internet gateway, but 10.20.0.0/16 (more specific than 0.0.0.0/0) is chosen for traffic to VPC B.

Table 8 — All historical routes in the three project route tables

Route table	Associated subnet	Destination	Target	Purpose	Evidence
ewb-rt-a-public rtb-074befa2c9ec0b2da	ewb-a-public	10.10.0.0/16	local	Intra-VPC A traffic (bastion to App-A)	Route list not captured; association E09; routes from draft/brief
ewb-rt-a-public rtb-074befa2c9ec0b2da	ewb-a-public	10.20.0.0/16	pcx-029955b9450e10c12	Bastion SSH to App-B through peering	Not captured (draft/brief); consistent with successful SSH to App-B (E03)
ewb-rt-a-public rtb-074befa2c9ec0b2da	ewb-a-public	0.0.0.0/0	igw-08dc44b91e4be2768	Bastion internet access / SSH entry	Not captured (draft/brief); IGW attachment E08
ewb-rt-a-private rtb-02fd4cd445c1b0910	ewb-a-private	10.10.0.0/16	local	Intra-VPC A traffic	E10
ewb-rt-a-private rtb-02fd4cd445c1b0910	ewb-a-private	10.20.0.0/16	pcx-029955b9450e10c12	App-A to VPC B (forward path)	E10 (status Active, origin: Create Route)
ewb-rt-b-private rtb-0b65c2d0934b9e333	ewb-b-private	10.20.0.0/16	local	Intra-VPC B traffic	E11
ewb-rt-b-private rtb-0b65c2d0934b9e333	ewb-b-private	10.10.0.0/16	pcx-029955b9450e10c12	App-B to VPC A (return path)	E11 (status Active, origin: Create Route)

Deleted route tables must not be read as current configuration: these are the routes that existed during testing.

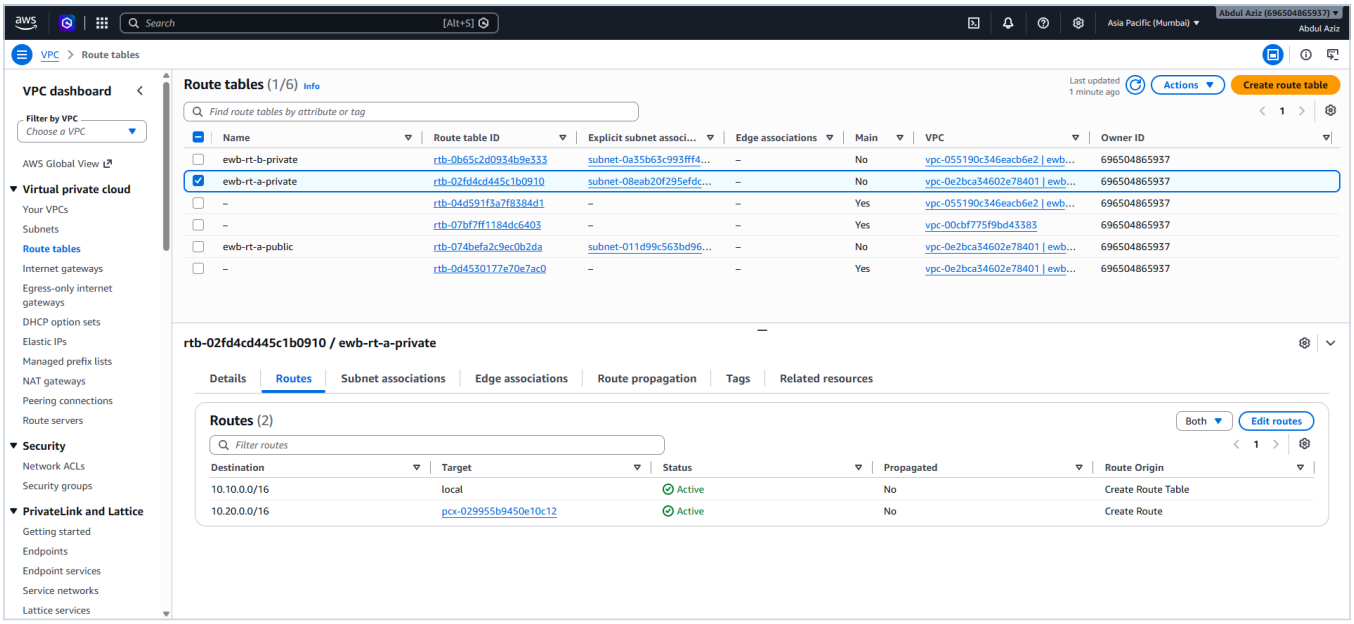


Figure 7 — Route Table ewb-rt-a-private: Local and Peering Route [E10]
Observed: rtb-02fd4cd445c1b0910 selected; Routes (2): 10.10.0.0/16 local; 10.20.0.0/16 → pcx-029955b9450e10c12, both Active
Demonstrates: App-A's subnet can send VPC B traffic into the peering connection (forward route)

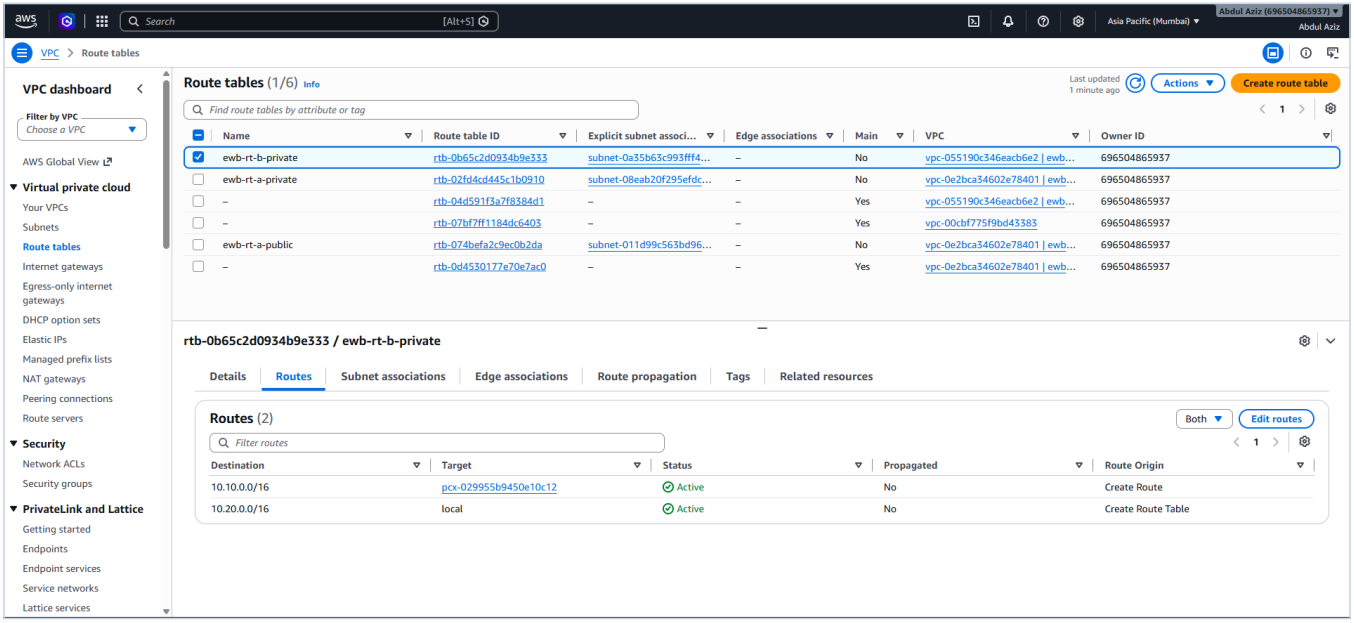


Figure 8 — Route Table ewb-rt-b-private: Return Route to VPC A [E11]
Observed: rtb-0b65c2d0934b9e333 selected; Routes (2): 10.10.0.0/16 → pcx-029955b9450e10c12; 10.20.0.0/16 local, both Active
Demonstrates: VPC B has a return route to VPC A; without it replies could not leave VPC B

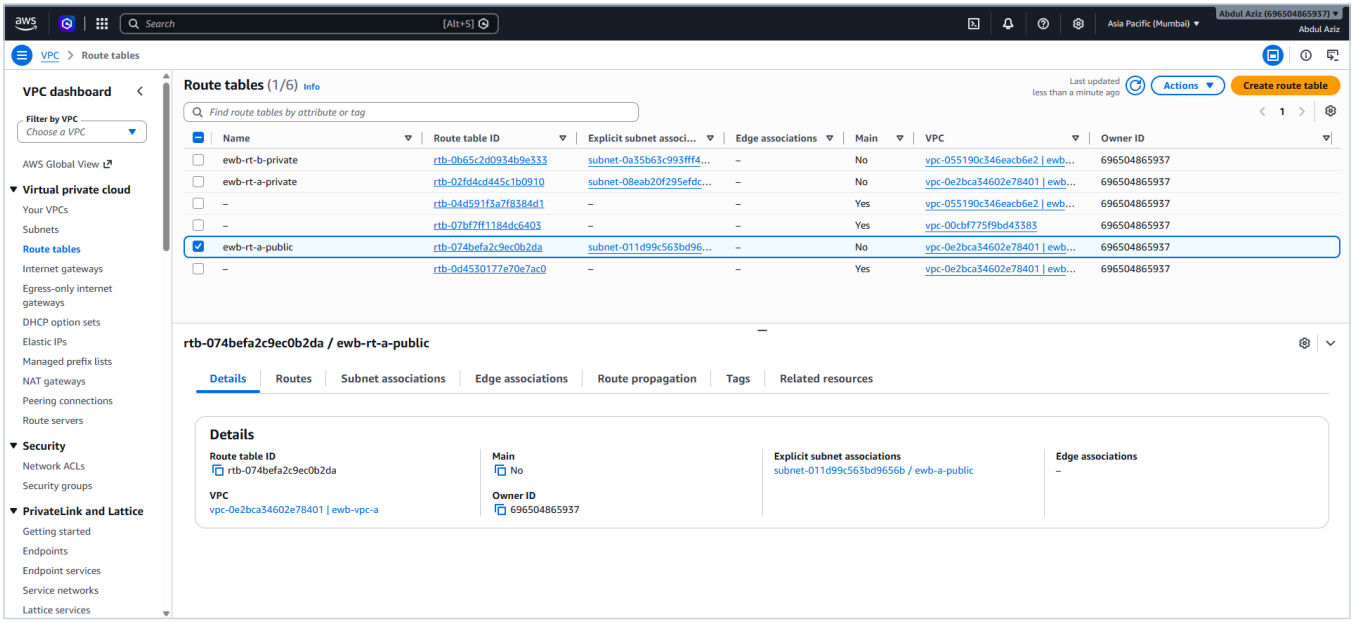


Figure 9 — Route Table ewb-rt-a-public: Details and Subnet Association [E09]
Observed: rtb-074befa2c9ec0b2da / ewb-rt-a-public, Main: No, explicit association subnet-011d99c563bd9656b / ewb-a-public, VPC ewb-vpc-a
Demonstrates: the public subnet is explicitly associated with its own (non-main) route table

8. Security-Group Inbound Rules

Table 9 — Inbound rules during testing (outbound left at the default allow-all)

Security group	Type / port	Source	Purpose	Evidence
ewb-sg-bastion sg-07efbf81da7c99333	SSH / TCP 22	`<redacted-admin-ip>/32`	Administrator public IP recorded during project (historical)	Draft/brief only; rule list not captured
ewb-sg-app-a sg-046c7ae629f8a628c	SSH / TCP 22	`10.10.1.10/32`	SSH only from bastion	E13 (sgr-0153583e1c5d0f4eb)
ewb-sg-app-a sg-046c7ae629f8a628c	All ICMP - IPv4 / ICMP / All	`10.20.1.10/32`	Pings initiated by App-B	E13 (sgr-0f4b585e0cc9a9082)
ewb-sg-app-b sg-081ad63f6378a6b76	All ICMP - IPv4 / ICMP / All	`10.10.2.10/32`	Pings initiated by App-A	Draft/brief only; rule list not captured
ewb-sg-app-b sg-081ad63f6378a6b76	SSH / TCP 22	`10.10.1.10/32`	SSH only from bastion (proxy source IP)	Draft/brief only; consistent with E03

Why the sources matter. A security group is a stateful, instance-level allow-list. Naming a single /32 source means only that host can open the permitted connection. The application servers accept SSH only from the bastion's private address, so a compromised or mistaken host elsewhere in either VPC cannot reach them on port 22. ICMP is accepted only from the one peer host. The bastion's rule is limited to the administrator's public address recorded at the time, which must not be read as the author's current IP. These restrictions are a sensible lab practice; they do not make this a production-grade or universally secure design (no logging, host hardening, Session Manager alternative, or network ACL design was demonstrated).

Deviation from the brief. The brief specifies Custom ICMP – IPv4 Echo request. E13 shows *All ICMP – IPv4* (ICMP, All). The rule is still scoped to one /32, but it permits more ICMP types than the brief asked for.

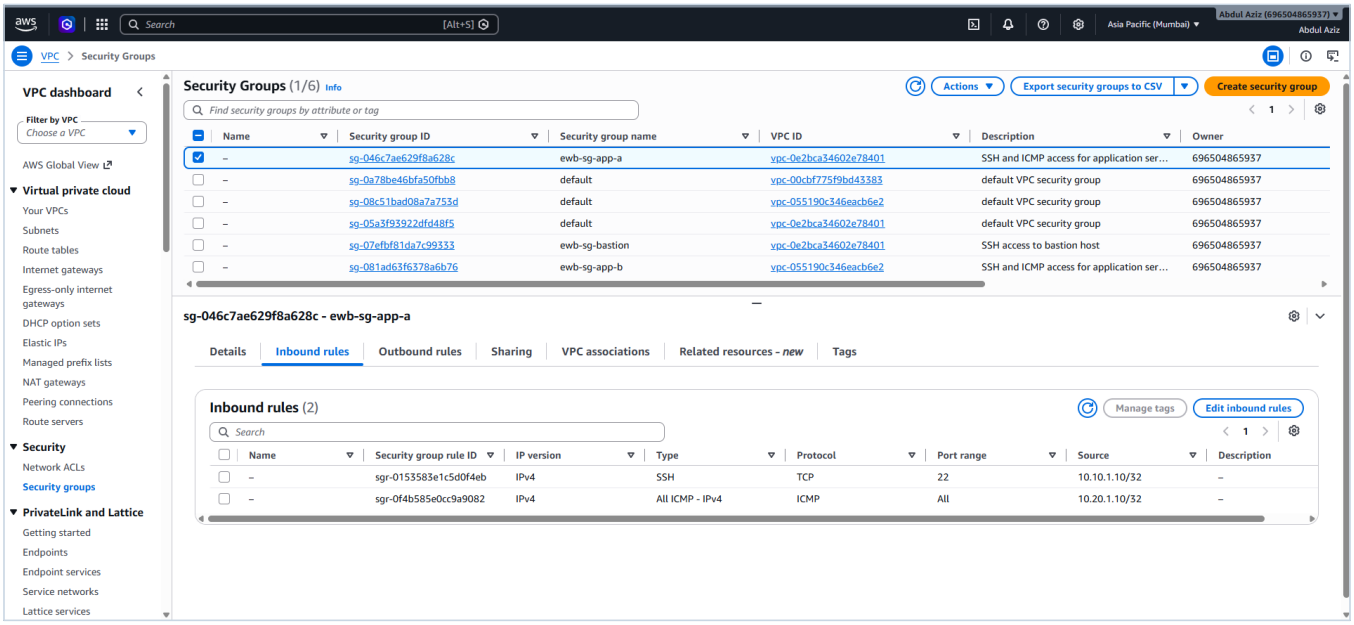


Figure 10 — Security Group ewb-sg-app-a: Inbound Rules [E13]
Observed: sg-046c7ae629f8a628c; SSH TCP 22 from 10.10.1.10/32 (sgr-0153583e1c5d0f4eb) and All ICMP – IPv4 from 10.20.1.10/32 (sgr-0f4b585e0cc9a9082); list also shows ewb-sg-bastion (VPC A) and ewb-sg-app-b (VPC B)
Demonstrates: App-A accepts SSH only from the bastion and ICMP only from App-B

9. Implementation Summary

The build followed the brief's sequence. Individual console steps were not captured, so this section records what was done and which later evidence confirms the result.

Table 10 — Implementation sequence

Step	Action	Confirmed by
1	Create ewb-vpc-a 10.10.0.0/16 and ewb-vpc-b 10.20.0.0/16 (VPC only, no IPv6, default tenancy)	E12
2	Create ewb-a-public, ewb-a-private (VPC A) and ewb-b-private (VPC B), all in ap-south-1a	E05–E07
3	Create ewb-igw-a and attach it to ewb-vpc-a	E08
4	Create ewb-rt-a-public (0.0.0.0/0 → IGW) and associate ewb-a-public; create ewb-rt-a-private and ewb-rt-b-private with explicit associations	E09–E11
5	Create security groups with source-restricted inbound rules	E13 (app-a); others D
6	Launch bastion (public IPv4) and two private instances with fixed private IPs	E05–E07
7	Request and accept peering ewb-a-b-peer; wait for Active	E12
8	Add peering routes: 10.20.0.0/16 in A-public and A-private; 10.10.0.0/16 in B-private	E10, E11; A-public D
9	Run tests, controlled failure, recovery; clean up	E01–E04; recovery and cleanup unevidenced

10. Access Path Through the Bastion Host

Administration follows **workstation → bastion (public IPv4, SSH from the administrator /32) → private instance (SSH from the bastion's private IP)**. The private key stays on the author's workstation; it is never copied to any instance. The captured commands use OpenSSH with `-o ProxyCommand="ssh ... -W %h:%p ec2-user@<bastion>"`, which performs the same bastion hop as the brief's ProxyJump. Because the bastion opens the onward connection, App-A and App-B see the **bastion's private IP 10.10.1.10** as the SSH source, which is why their rules use that address.

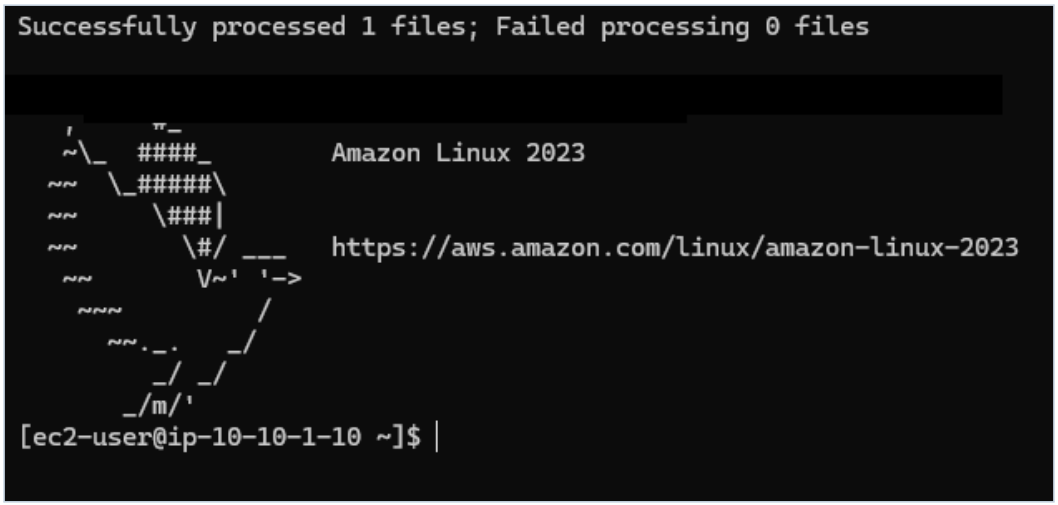


Figure 11 — SSH Access to the Bastion Host (ewb-bastion) [E01]
Observed: ssh ec2-user@[redacted public IPv4]; Amazon Linux 2023 banner; prompt ec2-user@ip-10-10-1-10
Demonstrates: administrator SSH works to the bastion's public address, and the instance is Amazon Linux 2023 at 10.10.1.10

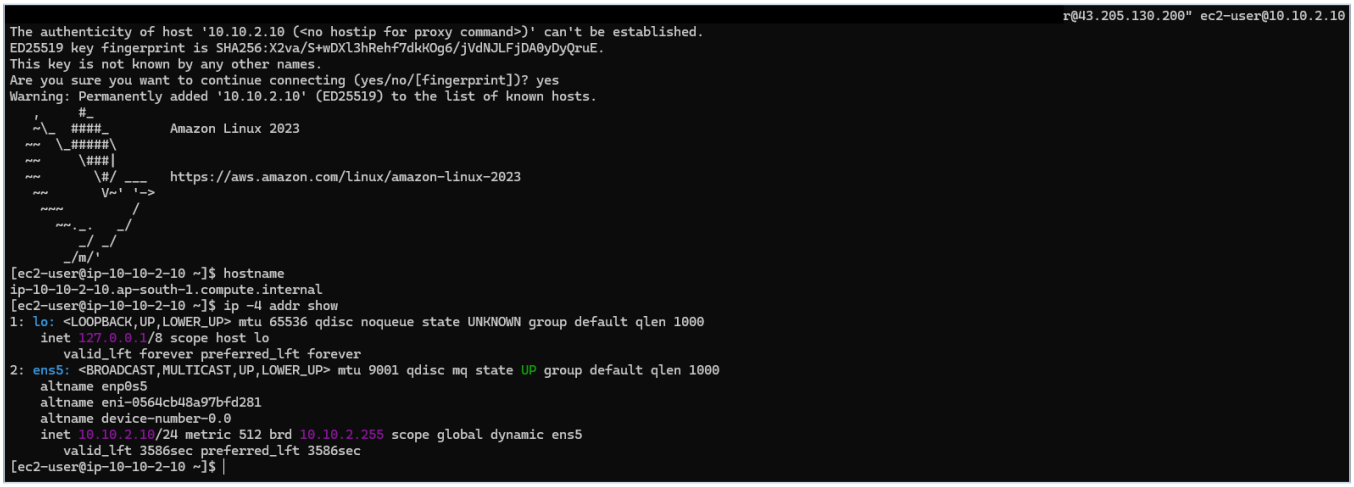


Figure 12 — SSH to ewb-app-a Through the Bastion [E02]
Observed: ProxyCommand via [redacted public IPv4] to 10.10.2.10; first-time host-key prompt accepted; hostname ip-10-10-2-10.ap-south-1.compute.internal; ens5 10.10.2.10/24
Demonstrates: the private instance is reachable only through the bastion and has the documented private address

11. Forward and Return Traffic Path

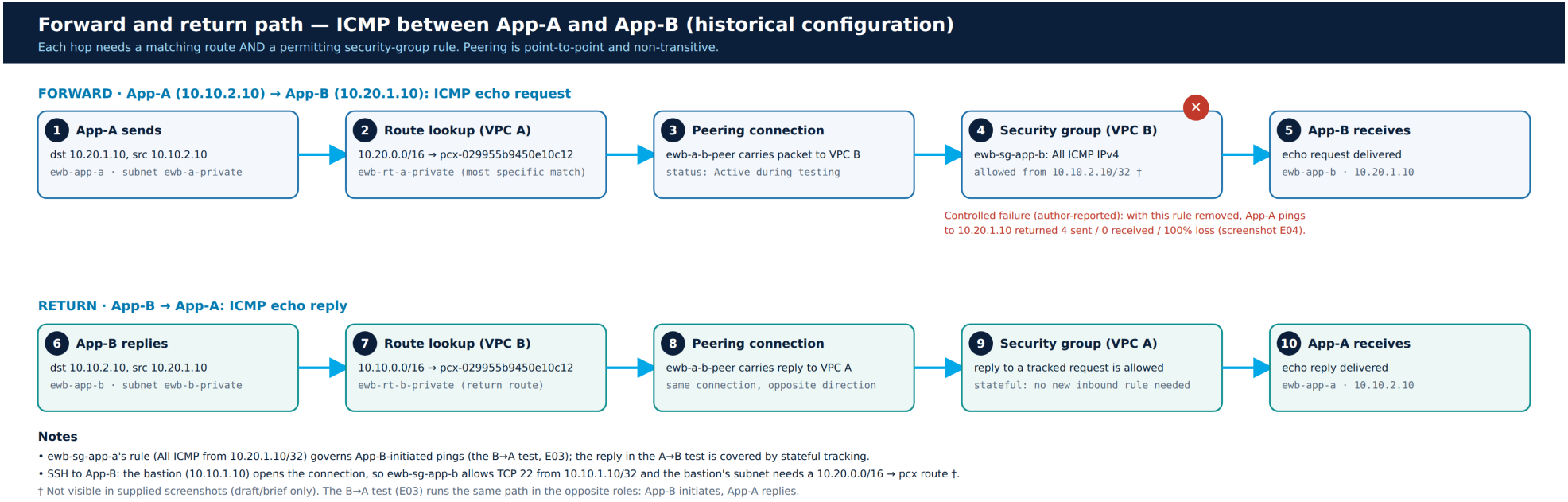


Figure 13 — Forward and return path for an App-A to App-B ping
Shows: ten hops: source, route lookup, peering, destination security group, delivery; then the reply in the reverse direction
Demonstrates: why both routes and a permitting rule are needed; the red marker is where the ICMP rule removal (E04) stopped the traffic

Forward (App-A → App-B). App-A sends to 10.20.1.10. The route table of its subnet matches 10.20.0.0/16 and forwards to the peering connection, which delivers the packet into VPC B. App-B's security group must allow ICMP from 10.10.2.10/32. **Return.** App-B answers 10.10.2.10; the route table of its subnet matches 10.10.0.0/16 and sends the reply back over the same peering connection. Because security groups are stateful, the reply to an allowed request is allowed without a new inbound rule on App-A. A route on only one side would deliver requests but lose every reply. Peering is point-to-point and non-transitive: it does not let VPC B use VPC A's Internet gateway.

12. Connectivity Test Matrix

Table 11 — Verification results (status reflects supplied evidence, not the author's recollection)

ID	Test	Command / action	Observed result	Evidence	Status
T1	SSH to bastion	ssh ec2-user@[redacted public IPv4] (key from local machine)	Login succeeded; prompt ip-10-10-1-10; Amazon Linux 2023	E01	**Verified**
T2	SSH to App-A via bastion	ssh with ProxyCommand through bastion to 10.10.2.10	Login succeeded; hostname ip-10-10-2-10.ap-south-1.compute.internal; ens5 10.10.2.10/24	E02	**Verified**
T3	Baseline before peering	ping -c 4 10.20.1.10 from App-A before peering/routes	Not captured. Expected result per brief: no replies	-	**Not captured**
T4	Forward ping App-A to App-B	ping -c 4 10.20.1.10 from App-A	Author reports 4/4 replies, 0% loss; no terminal capture supplied	-	**Reported**
T5	Reverse ping App-B to App-A	ping -c 4 10.10.2.10 from App-B	4 transmitted, 4 received, 0% packet loss; rtt avg 0.239 ms	E03	**Verified**
T6	SSH to App-B via bastion	ssh with ProxyCommand through bastion to 10.20.1.10	Login succeeded; prompt ip-10-20-1-10	E03	**Verified**
T7	Isolation of private hosts	EC2 instance details for App-A and App-B	Public IPv4 address '-' for both	E06, E07	**Verified**
T8	Controlled ICMP failure	ping -c 4 10.20.1.10 from App-A after App-B ICMP rule removed (author-reported action)	4 transmitted, 0 received, 100% packet loss	E04	**Failure shown**
T9	Recovery	Restore ICMP rule; retry ping	Author reports: first attempt immediately after restoring also failed; a later retry succeeded (4/4). No capture supplied	-	**Not captured**

A successful ping shows that a route, a peering path and a security rule worked for that direction at that moment. It does not prove every route in the design, and it says nothing about transitive routing, which peering does not provide.

13. Baseline Failure Before Peering

Before peering and peer routes exist, App-A has no valid path to 10.20.1.10, so the expected baseline is four packets sent with no replies.

14. App-A to App-B Connectivity Test

The project record reports 4 of 4 replies and 0% packet loss from App-A to 10.20.1.10. The controlled failure for the same command is shown in Figure 16 (E04).

15. App-B to App-A Connectivity Test



Figure 14 — App-B to App-A ICMP Test: Four Replies, Zero Packet Loss [E03]
Observed: ping -c 4 10.10.2.10 from ip-10-20-1-10: 4 transmitted, 4 received, 0% packet loss, time 3120 ms; rtt min/avg/max/mdev 0.194/0.239/0.302/0.039 ms; each reply ttl=127
Demonstrates: traffic from VPC B reaches VPC A privately, so VPC B's return route and App-A's ICMP rule (from 10.20.1.10/32) were working

16. App-B SSH Proof

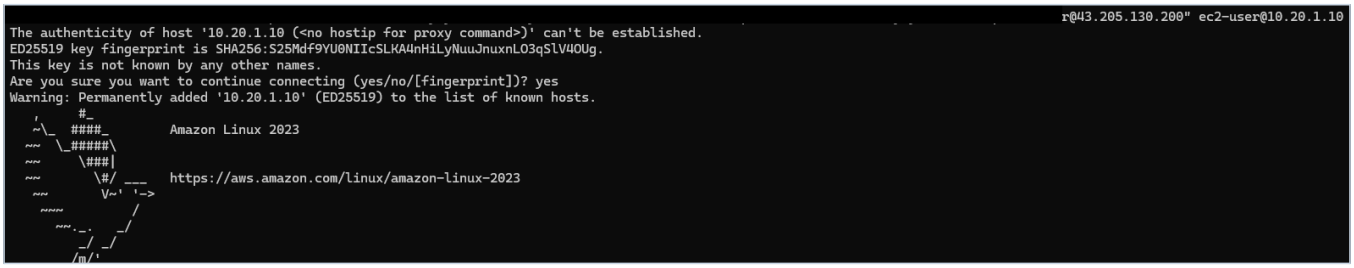


Figure 15 — SSH to ewb-app-b Through the Bastion [E03]
Observed: ProxyCommand via [redacted public IPv4] to 10.20.1.10 (host-key prompt accepted); Amazon Linux 2023 banner; prompt ec2-user@ip-10-20-1-10
Demonstrates: the bastion can reach App-B across the peering connection and App-B accepts SSH from the bastion's private IP

17. Controlled ICMP Failure

The author temporarily removed App-B's inbound ICMP rule (source 10.10.2.10/32) and repeated the ping from App-A. Replies stopped. According to the author only that rule was changed; routes and the peering connection were left in place. If so, the failure isolates the cause to the security group rather than routing, which shows that a route and an Active peering connection alone do not guarantee communication (the screenshot itself cannot show which setting was changed).

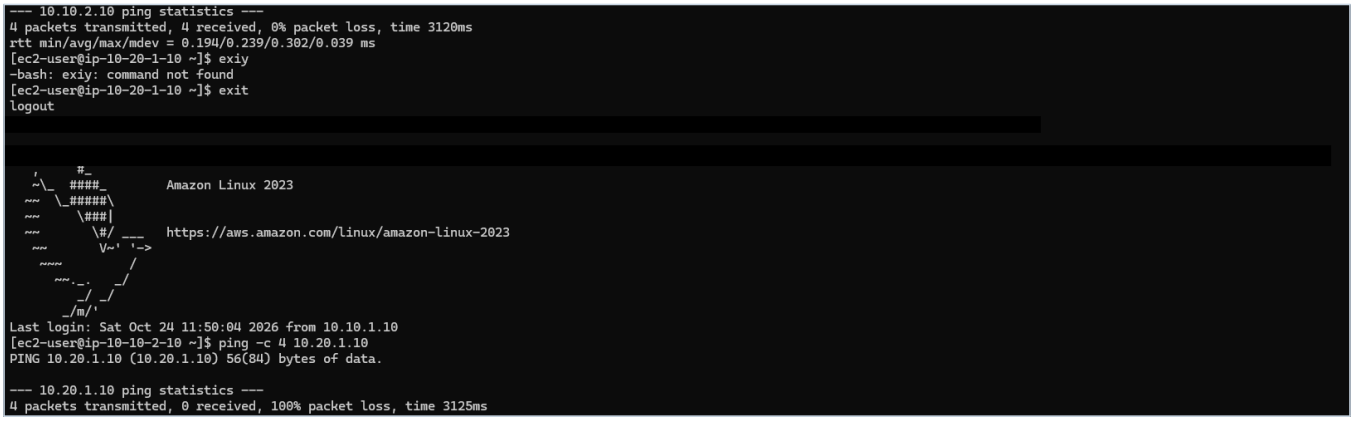


Figure 16 — Controlled ICMP Failure: No Replies After Rule Removal [E04]
Observed: tail of the successful B-to-A ping, a mistyped command ('exiy', command not found), exit, a new SSH session to App-A (Last login Sat Oct 24 11:50:04 2026 from 10.10.1.10), then ping -c 4 10.20.1.10: 4 transmitted, 0 received, 100% packet loss, time 3125 ms
Demonstrates: the A-to-B ICMP path stopped; consistent with the reported rule removal; also dates the testing to 24 Oct 2026 (instance clock)

18. Recovery and Explanation

- The ICMP rule (App-B inbound, source 10.10.2.10/32) was restored.
- An immediate ping from App-A still failed.
- A later retry succeeded with replies restored.

A short propagation delay for the security-group change is **one possible explanation**; it was not measured and is not proven. The brief itself advises waiting briefly and repeating the four-packet test, which is why the retry matters: a single immediate failure after a change is not conclusive.

19. Cleanup and Final Resource State

The author reports that the three instances were terminated and that the project subnets, peering connection, both VPCs and the Internet gateway were deleted. Cleanup was performed after evidence collection to avoid ongoing EC2, EBS and public IPv4 charges, and to leave no internet-facing host. The final states below reflect the author's cleanup report.

Table 12 — Cleanup record

Resource category	Historical purpose	Final status	Evidence
EC2 instances (ewb-bastion, ewb-app-a, ewb-app-b)	Bastion host and two private application servers	Terminated (reported)	None supplied
Subnets (ewb-a-public, ewb-a-private, ewb-b-private)	Address segments for the three instances	Deleted (reported)	None supplied
VPC peering connection ewb-a-b-peer	Private link between VPC A and VPC B	Deleted (reported)	None supplied
Internet gateway ewb-igw-a	Internet entry for the bastion in VPC A	Deleted (reported)	None supplied
VPCs ewb-vpc-a, ewb-vpc-b	Isolated network containers	Deleted (reported)	None supplied
Route tables (3 custom)	Routing for each subnet	Not individually evidenced; custom route tables cannot outlive their VPC, so deleting the VPCs removes them (AWS behaviour, not observed here)	None supplied
Security groups (3 custom)	Instance-level inbound filtering	Not individually evidenced; removed with their VPCs (same caveat)	None supplied
Key pair ewb-lab, EBS volumes, Elastic IPs, tags	Access key; root volumes; none created	Not stated by the author; NEEDS REVIEW	None supplied

Table 13 — Three states kept distinct

State	What it means	Source
Historical lab configuration	The resources and routes defined in Sections 5–8, as they existed during testing	Brief, draft, screenshots
Observed test state	Only what the screenshots E01–E13 show at capture time (Peering Active, instances Running)	Screenshots
Post-cleanup state	Project resources deleted; the account's default VPC resources seen in screenshots were never part of the project	Author report; no evidence

Evidence and Verification Notes

Console screenshots directly support the Active peering state, both private-subnet peering routes, the VPC A Internet Gateway attachment, private-only application instances, SSH access through the bastion, App-B-to-App-A ping success, and the controlled App-A-to-App-B ping failure. The App-A-to-App-B success, pre-peering baseline failure, recovery after restoring the ICMP rule, and cleanup actions are reported in the project record but do not have corresponding screenshots in the supplied set. The public route-table route list and all security-group inbound rules were not fully captured. These distinctions are preserved in the test summary and evidence index; no missing evidence has been reconstructed.

20. Lessons Learned and Troubleshooting

Table 14 — Problems, causes and outcomes

Observation	Confirmed cause?	How to diagnose	Outcome
Ping fails although peering is Active	Confirmed in the controlled test: security-group rule removed (author-reported)	Check destination SG inbound rule, then both route tables, then host IPs	Replies stop; restored when the rule returned
Immediate retry fails after restoring the rule	Not confirmed. Rule-propagation delay is plausible	Wait briefly and repeat the four-packet test	A later retry succeeded (reported)
Typing error ('exiy') in the terminal	Confirmed in E04: command not found; harmless	None needed	Session continued
App-B SSH needs a route from the bastion's subnet	Inferred, not captured: A-public route to 10.20.0.0/16 (E03 succeeded)	Check ewb-rt-a-public and App-B SG source (bastion private IP)	SSH to App-B succeeded

- VPC peering is a link, not a merged network: both sides need routes and the destination needs a permitting rule.
- Routes are selected by longest prefix, and each subnet is governed by its own associated route table.
- Security groups are stateful, so inbound rules matter for the direction that *starts* a connection.
- A bastion lets private servers stay private; the proxied SSH source is the bastion's private IP.

- Capture evidence at every checkpoint before cleanup; the baseline, forward-success, recovery and cleanup captures were not available for this report.

21. Design Boundaries and Future Improvements

- Baseline, forward-success, recovery and cleanup captures are not present in the supplied evidence set; reported outcomes are distinguished from screenshot-supported results in Appendix A.
- Single AZ, single instance per role, t3.micro: no resilience was tested, and no performance claim is made.
- ICMP rule is broader than the brief's Echo-request type (Section 8); tighten to type 8.
- No VPC Flow Logs, so packet acceptance or rejection was inferred from ping output rather than observed in logs.
- Possible extensions: narrower peering routes (specific subnets), SSM Session Manager instead of a public bastion, Flow Logs, infrastructure-as-code, and a scripted test that captures timestamps.

22. Appendix A — Evidence Index

Table 15 — Screenshot evidence index

ID	File (repository)	Shows	Supports
E01	connectivity/E01_bastion_ssh_login.png	Windows terminal: ssh as ec2-user to [redacted public IPv4]; Amazon Linux 2023 banner; prompt ec2-user@ip-10-10-1-10	SSH to bastion; AL2023; bastion private IP 10.10.1.10
E02	connectivity/E02_app-a_ssh_via_bastion.png	SSH to 10.10.2.10 using ProxyCommand through bastion; host-key prompt; hostname ip-10-10-2-10; ip -4 addr shows 10.10.2.10/24	SSH to App-A via bastion; App-A private IP and /24 prefix
E03	connectivity/E03_app-b_ssh_via_bastion_and_ping_to_app-a.png	SSH to 10.20.1.10 via bastion; prompt ip-10-20-1-10; ping -c 4 10.10.2.10: 4 transmitted, 4 received, 0% loss, rtt 0.194/0.239/0.302/0.039 ms	App-B SSH proof; B-to-A ping success
E04	failure-recovery/E04_app-a_ping_to_app-b_100pct_loss.png	Tail of the E03 ping, a typo ('exiy') then exit; new SSH to 10.10.2.10 (Last login Sat Oct 24 11:50:04 2026 from 10.10.1.10); ping -c 4 10.20.1.10: 4 transmitted, 0 received, 100% loss	ICMP failure from App-A to App-B; test date (24 Oct 2026)
E05	compute/E05_ec2_ewb-bastion_details.png	EC2 list (3 instances Running, 3/3 checks, t3.micro, ap-south-1a) with ewb-bastion selected: public IPv4 [redacted public IPv4], private 10.10.1.10, VPC ewb-vpc-a, subnet ewb-a-public	Bastion inventory; public IPv4; instance IDs
E06	compute/E06_ec2_ewb-app-a_details.png	ewb-app-a selected: i-03f2e1cdcc22ab028, no public IPv4, private 10.10.2.10, VPC ewb-vpc-a, subnet ewb-a-private	App-A private-only; inventory
E07	compute/E07_ec2_ewb-app-b_details.png	ewb-app-b selected: i-09e1acf311b10836d, no public IPv4, private 10.20.1.10, VPC vpc-055190c346eacb6e2, subnet ewb-b-private	App-B private-only; inventory
E08	networking/E08_internet_gateway_ewb-igw-a.png	Internet gateways list (2): ewb-igw-a igw-08dc44b91e4be2768 Attached to vpc-0e2bca34602e78401 (ewb-vpc-a); second IGW igw-09386cdb394e61456 attached to default VPC vpc-00cbf775f9bd43383	IGW attached to VPC A only (among project VPCs)
E09	networking/E09_route_table_ewb-rt-a-public_details-tab.png	Route tables list (6) with ewb-rt-a-public selected; Details tab: rtb-074befa2c9ec0b2da, Main: No, explicit association subnet-011d99c563bd9656b / ewb-a-public	Public route table and its subnet association
E10	networking/E10_route_table_ewb-rt-a-private_routes.png	ewb-rt-a-private rtb-02fd4cd445c1b0910 selected; Routes (2): 10.10.0.0/16 local; 10.20.0.0/16 pcx-029955b9450e10c12, both Active	App-A forward route to VPC B
E11	networking/E11_route_table_ewb-rt-b-private_routes.png	ewb-rt-b-private rtb-0b65c2d0934b9e333 selected; Routes (2): 10.10.0.0/16 pcx-029955b9450e10c12; 10.20.0.0/16 local, both Active	Return route from VPC B to VPC A
E12	networking/E12_vpc_peering_ewb-a-b-peer_active.png	Peering connections (1/1): ewb-a-b-peer pcx-029955b9450e10c12 Active; requester vpc-0e2bca34602e78401/ewb-vpc-a 10.10.0.0/16; accepter vpc-055190c346eacb6e2/ewb-vpc-b 10.20.0.0/16; both Mumbai	Active peering; VPC IDs and CIDRs
E13	security/E13_sg_ewb-sg-app-a_inbound_rules.png	Security groups list (6) with ewb-sg-app-a selected; Inbound rules (2): SSH TCP 22 from 10.10.1.10/32; All ICMP IPv4 from 10.20.1.10/32	App-A rules; SG IDs and VPC membership of all three project SGs
REF-A	original diagram	Author's earlier diagram	Reference only (not evidence of configuration)

23. Appendix B — Discrepancy Log

Table 16 — Conflicts found and how each was handled

#	Where	Conflict	Resolution
D1	Original diagram vs screenshots	Original diagram lists VPC A/B as vpc-0a1234567890abcd / vpc-00cbf775f9bd43383; peering pcx-0abc1234; IGW igw-0def5678; placeholder subnet and instance IDs → Screenshots show vpc-0e2bca34602e78401, vpc-055190c346eacb6e2, pcx-029955b9450e10c12, igw-08dc44b91e4be2768 and the subnet/instance IDs in the inventory. vpc-00cbf775f9bd43383 is the default VPC	Resolved: original diagram not used; new diagram built from screenshot values
D2	Original diagram vs brief/draft	Original shows ewb-sg-bastion SSH from 0.0.0.0/0 → Brief requires your /32; draft records /32	Resolved in favour of brief/draft; bastion rule remains unverified by screenshot
D3	Original diagram	Shows 0.0.0.0/0 '(No route)' row in rt-b-private and omits ewb-rt-a-private; private subnet IDs differ from the screenshots → E10/E11/E06/E07	Resolved: corrected in new diagram
D4	Draft report section 7	States App-A's SG ICMP rule permits the return path of the A-to-B ping → Security groups are stateful; replies to permitted traffic are allowed automatically (brief p.6). App-A's ICMP rule governs pings that App-B initiates	Resolved: explanation corrected
D5	Draft report	Marks forward ping, recovery and controlled failure as 'screenshot saved by user' → Only the failure (E04) and the reverse ping (E03) were supplied	Unresolved: forward-ping and recovery screenshots not supplied
D6	Draft report	Cleanup 'pending'; instances running → Author now reports all cleanup done	Unresolved: no cleanup evidence; report states cleanup as reported
D7	Brief p.6 vs screenshot E13	Brief specifies Custom ICMP IPv4 Echo request (type 8) → E13 shows 'All ICMP - IPv4' (ICMP, All) scoped to a /32	Open deviation: broader ICMP type than the brief; still restricted to one source host
D8	Brief p.8 vs evidence	Brief shows SSH config with ProxyJump → Screenshots use ssh -o ProxyCommand=... -W %h:%p	Resolved: functionally equivalent bastion hop; ProxyCommand documented
D9	File name rt-a-public_routes.png	Name implies a Routes tab → Image shows the Details tab only	Resolved: renamed E09 ..._details-tab; public routes marked unverified
D10	Brief p.2 baseline	Requires a failed test before peering → No baseline capture supplied	Unresolved: requirement not captured
D11	Draft: 'Last updated' / date	Draft says completion pending → E04 shows 'Last login Sat Oct 24 11:50:04 2026' from the instance clock	Resolved: testing date 24 Oct 2026 (instance clock, time zone not shown); completion/cleanup date not evidenced
D12	Screenshots E08-E13	Lists include default-VPC IGW, route tables and security groups → These are not project resources	Resolved: excluded from the project inventory; listed separately

24. References

- Amazon VPC Peering Guide — What is VPC peering? https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html
- Amazon VPC Peering Guide — Create a VPC peering connection. https://docs.aws.amazon.com/vpc/latest/peering/create-vpc-peering-connection.html
- Amazon VPC Peering Guide — Update your route tables for a VPC peering connection. https://docs.aws.amazon.com/vpc/latest/peering/vpc-peering-routing.html
- Amazon VPC Peering Guide — Configurations with routes to an entire VPC. https://docs.aws.amazon.com/vpc/latest/peering/peering-configurations-full-access.html
- Amazon VPC User Guide — Security groups. https://docs.aws.amazon.com/vpc/latest/userguide/vpc-security-groups.html
- EWB Courses — AWS VPC Peering project brief (Attachment A).

These documentation URLs were located through search results while preparing the report; they were not each opened in full, and the security-group URL was not retrieved at all. Documentation supports the explanations; project facts come only from the supplied evidence.